

CampusAssetIQ — System Architecture & Delivery Report

Enterprise Hardware Asset Intelligence, QR Lifecycle & Incident Management Platform

Target Institution: Swaminarayan University & PSM Hospital	Production Status: LIVE (24/7 Global Cloud)
Recipient: Head of Department (HOD) — Information Technology	Cloud Endpoint: https://it-support-6np1.onrender.com
Document Scope: Architectural Overview, Functional Modules & Deployment	Custom Domain Target: support.swaminarayanuniversity.ac.in

1. Executive Summary

CampusAssetIQ is an enterprise-grade hardware lifecycle tracking, preventive maintenance, and real-time incident intelligence system engineered specifically for the mission-critical operations of **PSM Hospital** and **Swaminarayan University**. The platform modernizes legacy, disconnected Excel spreadsheets into a centralized, relational PostgreSQL database architecture. It provides high-speed hardware registry management, physical ISO-standard QR asset tagging, instant smartphone incident triage for clinical staff, equipment breakdown lifecycle logging, and preventive maintenance compliance tracking.

The system has been completely containerized, hardened for security, and deployed to a **24/7 High-Availability Cloud Infrastructure** with active HTTPS SSL certification, automated database migrations, and CI/CD version control integration.

2. Technology Stack & Architecture

Layer	Technology	Architectural Rationale & Implementation Details
Backend Core	Python 3.12 / Django 6.1 (MTV)	Server-rendered architecture providing sub-millisecond page loads with zero Single-Page-App (SPA) runtime bloat and memory leaks.
Primary Database	PostgreSQL 16 (Relational Engine)	ACID-compliant relational database ensuring 100% data integrity, foreign key constraints, and zero risk of SQLite lock contention.
Application Server	Gunicorn WSGI (4 Workers)	High-concurrency production WSGI server handling simultaneous requests from hospital wards, labs, and staff portals.
Static Assets	WhiteNoise Engine	Gzip/Brotli compressed manifest storage serving CSS, JS, and Lucide vector icons directly from memory with HTTP/2 caching.
Frontend UI/UX	Tailwind CSS + Vanilla ES6+	Lightweight, reactive client interfaces with zero external framework dependencies; native HTML5 QR video stream decoding.
Cloud Hosting	Render Cloud PaaS (Singapore Edge)	Sub-50ms latency connection to Western India; 24/7 uptime guarantee, automatic process supervisor, and SSL termination.

3. Core Engineering Innovations & Structural Improvements

3.1 Single Identifier Architecture (Zero 'System Name')

Historically, asset tracking suffered from dual-identity confusion (e.g., matching ambiguous names like 'ICU-BED-MON-01' with hardware serials). We permanently re-engineered the schema to enforce **asset_id** (e.g. **PSM/IT/2F/C-201**) as the **absolute single source of truth** across all database models, APIs, templates, custody transfer logs, and physical QR codes. The redundant `system_name` attribute has been permanently purged.

3.2 Automated Inventory → Asset Tag Center Synchronization

A real-time database synchronization pipeline was engineered between the Inventory Registry and the Asset Tag Printing Center. When an administrator registers a hardware device (or imports in bulk), the system automatically executes `POST /api/devices/save/`, persists the record into PostgreSQL, and prepends the newly created asset sticker as **Card #1** in the Tag Center.

3.3 Strict 'Required-Data-Only' Filtering for Physical Asset Stickers

Physical hardware stickers must be legible from a distance and free from cognitive clutter. In accordance with clinical ISO standards, internal machine specs (RAM chip speeds, processor buses, IP addresses, MAC addresses) are strictly excluded from the physical sticker. **Only vital physical operational fields are transferred:** Institution Brand, Asset ID, Device Category Badge, Assigned Custodian, Physical Room Location, Hardware Serial Number, and a High-Density Vector QR Matrix.

3.4 High-Density Vector QR Generation Engine

Rather than relying on low-resolution raster images, the system incorporates an in-browser ISO-18004 compliant vector SVG QR generator. QR codes are dynamically compiled with Error Correction Level 'M', enabling instant recognition by any iOS or Android camera even on curved surfaces, medical carts, or low-light ICU environments.

4. Functional Modules & Capabilities

Module	Key Features & Engineering Highlights
Hardware Inventory Registry /inventory/	- Categorized asset tracking across 5 standard hardware classes: CPU, Display, Keyboard, Mouse, and Printer. - Real-time text search across Asset ID, Serial, Doctor Name, and IP address. 15-Column bulk Excel import/export parser with duplicate detection and automatic floor allocation.
Location & Spatial Explorer /location/	- Hierarchical 3-tier architectural mapping: Building → Floor → Department / Room. - Multi-tenant institutional scoping: PSM Hospital (Clinical) vs. Swaminarayan University (Academic). - Dynamic room capacity indicators and localized hardware count rollups.
Asset Tag & QR Center /tag/	- Dedicated high-density sticker grid with live printable A4 portrait sheet rendering. 1-Click single-tag instant printer dialog with zero browser margin bleed. - Real-time search and filter by Asset ID, Category, or Custodian.
Mobile Incident Intelligence /report/<asset_id>/	- Zero-login direct smartphone camera complaint logging for doctors and nursing staff. - Automatic hardware context detection (Asset ID, room location, assigned user pre-filled). - Category selection (Hardware, Display, Network, OS, Peripherals) with priority urgency flags.
Breakdown & Downtime Register /breakdown-register/	- Comprehensive audit register tracking Mean Time to Repair (MTTR) and Mean Time Between Failures (MTBF). - Status lifecycle: Open / Triage → Technician Assigned → In Repair → Resolved & Backfilled. - Location column tracking exact clinical room where breakdown occurred.
PMS Schedule Tracker /pms-schedule/	- Preventive maintenance scheduling for high-availability medical and lab equipment. - Cycle frequencies: Quarterly, Half-Yearly, Annually, Monthly with vendor contract tracking. - Status tracking: Completed, Pending 4th Cycle, Due Soon, Overdue compliance warnings.
Dual Authentication Portals /admin-login/ & /portal/	Admin Command Portal: Obsidian cyber-palette matching mission-critical operations. Staff Self-Service Portal: Clean healthcare clinical theme for staff self-service and ticket history.

5. Cloud Infrastructure, Scalability & Capacity

The system is deployed on high-availability managed cloud architecture with dedicated compute and persistent storage:

Metric	Allocated Capacity	Real-World Operational Impact (PSM Hospital & University)
Database Storage	1 GB (1,000 MB) PostgreSQL	Holds over 2,000,000 (20 Lakh) equipment & ticket records . At an average campus consumption rate of ~10 MB/year, this provides 50–100 years of continuous operational capacity .
Monthly Bandwidth	100 GB / month	Supports over 1,000,000 page views per month with sub-second response times across campus Wi-Fi and external mobile networks.
Compute Runtime	512 MB High-Performance RAM	Zero-memory-leak Python 3.12 WSGI environment with automated worker recycling and health-check supervisor.
SSL / Security	Cloudflare Managed TLS 1.3	Valid green padlock HTTPS certificate, enabling HTML5 navigator.mediaDevices.getUserMedia camera permissions on all mobile browsers.

CI/CD DevOps	GitHub Webhook Deployment	Every code commit pushed to krish-228/campus-asset-iq automatically triggers test compilation, static collection, and live zero-downtime deployment in ~60 seconds.
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6. Institutional Domain Integration Roadmap

To provide a branded university presence, the application can be seamlessly mapped to the official subdomain **support.swaminarayanuniversity.ac.in** without requiring any risky nameserver migrations or server disruptions:

Step	Action Item	Technical Execution Details
Step 1	Render Custom Domain Binding	In Render Web Service Settings → Custom Domains, bind support.swaminarayanuniversity.ac.in. Render automatically provisions the Let's Encrypt SSL certificate.
Step 2	University DNS CNAME Record	In the University's primary DNS zone editor (e.g. iredtech.biz), add a standard CNAME record: • Host / Name: support • Type: CNAME • Value / Destination: it-support-6np1.onrender.com • TTL: 14400 (or Automatic)
Step 3	Zero-Disruption Cutover	The university's main website (swaminarayanuniversity.ac.in) and institutional mail servers remain 100% untouched. All traffic to support... automatically routes to the CampusAssetIQ cloud instance.

7. Engineering Sign-Off & Recommendations

Next Recommended Steps for Institutional Rollout:

1. **Physical QR Sticker Rollout:** Print the standard A4 tag sheets from the Asset Tag Center and apply vinyl stickers to existing computers, monitors, and ICU terminals.
2. **DNS Record Addition:** Coordinate with the network administrator to insert the support CNAME pointer in the university DNS manager.
3. **Staff Orientation:** Conduct a brief 10-minute briefing showing hospital nursing supervisors and lab technicians how to scan the QR stickers using standard phone cameras.

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Date: September 18, 2026	Signature: _____